8. Father of a family wishes to divide his square field bounded by $x=0$, $x=4$, $y=4$ and $y=0$ along the curve $y^2=4x$ and $x^2=4y$ into three equal parts for his wife, daughter and son. Is it possible to divide? If so, find the area to be divided among them.
9. The curve $y=(x-2)^2+1$ has a minimum point at P. A point Q on the curve is such that the slope of PQ is 2. Find the area bounded by the curve and the chord PQ.
10. Find the area of the region common to the circle $x^2+y^2=16$ and the parabola $y^2=6x$.

9.9 Volume of a solid obtained by revolving area about an axis

Definite integrals have applications in finding volumes of solids of revolution about a fixed axis. By a solid of revolution about a fixed axis, we mean that a solid is generated when a plane region in a given plane undergoes one full revolution about a fixed axis in the plane. For instance, consider the semi circular plane region inside the circle $x^2+y^2=a^2$ and above the x-axis. See Fig.9.34.

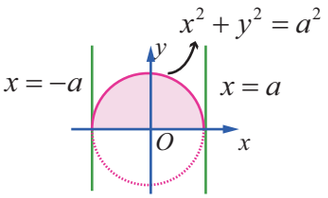


Fig. 9.34

If this region is given one complete rotation (revolution for $360°=2\pi$ radians) about x-axis, then a solid called a sphere is generated.

In the same manner, if you want to generate a right-circular cylinder with radius a and height h, you can consider the rectangular plane region bounded between the straight lines $y=0$, $y=a$, $x=0$ and $x=h$ in the xy-plane. See Fig.9.35. If this region is given one complete rotation (revolution for $360°=2\pi$ radians) about x-axis, then a solid called a cylinder is generated.

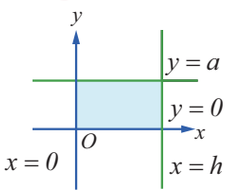


Fig. 9.35

We restrict ourselves to obtain volume of solid of revolution about x-axis or y-axis. Whenever solid of revolution about x-axis is considered, the plane region that is revolved about x-axis lies above the x-axis. So, in this region $y\geq 0$. Whenever solid of revolution about y-axis is considered, the plane region that is revolved about y-axis lies to the right of y-axis. So, in this region $x\geq 0$. We shall find the formula for finding the volume of the solid of revolution of the plane region in the first quadrant bounded by the curve $y=f(x)$, x-axis and the lines $x=a$ and $x=b>a$ about x-axis. **The derivation of the formula is based upon the formula that the volume of a cylinder of radius r and the height h is $\pi r^2 h$.**

Assume that every line parallel to y-axis lying between the lines $x=a$ and $x=b>a$ cuts the curve $y=f(x)$ in the first quadrant exactly at one point. Divide $[a,b]$ into n segments by $x_1, x_2, ..., x_{n-1}$ such that

$$a=x_0<x_1<x_2<...<x_{n-1}<x_n=b,\ x_i-x_{i-1}=\Delta x=\frac{b-a}{n},\ i=1,2,...,n.$$

For each $i=0,1,2,...,n-1$, the region in the xy-plane between the ordinates at x_i and $x_i+\Delta x$ which lies between the x-axis and the curve $y=f(x)$ can be **approximated** to an infinitesimal rectangle having area $y_i\Delta x$, where $y_i=f(x_i)$. When the plane region bounded by the curve $y=f(x)$, x-axis, and lines $x=a$ and $x=b$ is rotated by $360°$ about x-axis, each of the infinitesimal rectangles at $x=x_i$ also revolves and generates an elementary solid which is **approximately** a thin cylindrical disc with radius y_i and height Δx. See Fig.9.36. The volume of the cylindrical disc at

12th_Maths_Vol 2_EM_CH 9_App of Integration_17-02-20.indd 135 16-12-2022 16:42:05